Mask-Enhanced Contrastive Learning for Hyperspectral Image Classification

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***Abstract*— Recently, self-supervised learning (SSL) has gained great prominence in hyperspectral image classification (HSIC) due to its powerful capability to alleviate data-hunger problem. The generative-based method and the contrastive-based method have become two main streams in the field of SSL. To fully integrate the merits of both of them, we propose an efficient hybrid SSL method, that is, mask-enhanced contrastive learning (MECL). Essentially, MECL remains a prototypical contrastive-learning (CL) method, but incorporates the masking-and-predicting idea of the generative-based method. When the sample gradually matches the prototypes, feature reconstruction is implicitly performed in MECL step by step. Furthermore, we design a spatial–spectral multimasking mechanism for hyperspectral data and also propose two complementary strategies to prevent MECL from collapsing. To demonstrate the performance of MECL, a wide range of experiments are carried out in our work. On the one hand, the results of ablation experiments prove that HSIC does benefit from the fusion of SSL methods. On the other hand, the classification results on three hyperspectral datasets confirm that our MECL is an effective SSL method with high stability and strong reliability.**

***Index Terms*— Contrastive learning (CL), hyperspectral image classification (HSIC), masking, self-supervised learning (SSL).**

I. Introduction

HYPERSPECTRAL remote sensing image is a multidimensional information acquisition technology that combines imaging technology and spectroscopy technology [1]. Hyperspectral images (HSIs) are composed of hundreds of continuous spectral bands and contain a wealth of spatial and spectral information, providing bread-and-butter support for HSIC. Importantly, accurate classification results are a prerequisite for various applications, such as pollution monitoring, medical diagnosis, agricultural assessment, and geological exploration [2].

In the past decades, researchers have designed numerous handcrafted traditional machine learning algorithms, which have achieved remarkable success in HSIC, such as the support vector machine (SVM) [3], random forest (RF) [4], K-nearest neighbor (KNN) [5], and so on. However, these methods based on shallow feature extraction technologies always demand careful engineering and domain knowledge of experts, which usually limits their practicality. Deep learning algorithms, able to automatically learn advanced features from raw data, have been widely utilized in HSIC. They exhibit outstanding performance because of their strong robustness and adaptability. Autoencoders (AEs) [6], [7], convolutional neural networks (CNNs) [8], [9], [10], [11], recurrent neural networks (RNNs) [12], and graph convolutional networks (GCNs) [13], [14] are the most commonly used networks.

The stacked autoencoder (SAE) [6] generated by stacking multiple AEs is probably the first deep learning algorithm to be employed for HSIC. Similar to most traditional machine learning algorithms, the SAE is also a spectrum classifier that pays little attention to the spatial dependencies of adjacent pixels. By contrast, the convolutional autoencoder (CAE) [7] takes full advantage of both spatial and spectral information. For this reason, it performs better than the SAE. Nevertheless, it focuses on all features, regardless of whether they are important or not. To alleviate this problem, Pande and Banerjee [7] add an attention module to the CAE, so that the autoencoder is driven to consider the relative importance of features when converting high-dimensional features to low-dimensional features. CNNs are the most widely used networks in the domain of HSIC, benefiting from local connectivity and parameter sharing. As the most common form of CNNs, 2D-CNNs [8], [9] perceive local features in the spatial dimension of HSI data and synthesize local features at a high level to obtain global information. Unfortunately, 2D-CNNs fail to extract distinguishing features from the spectral dimension. Unlike 2D-CNNs, 3D-CNNs [10], [11] are good at extracting the joint spatial–spectral information of HSI data. For example, SSRN [11] exploits the residual blocks to connect each 3-D convolution layer to facilitate the gradient backpropagation, resulting in superior performance across multiple HSI datasets. But, in most cases, there are always substantial parameters in a 3D-CNN model, which naturally leads to high-computational burden. To make full use of the advantages of 2D-CNNs and 3D-CNNs, hybrid CNNs combine them while allowing them to overcome the problems of each other to some extent. For example, the classical HybridSN [15] utilizes 3D-CNN to derive joint spatial–spectral representations and uses 2D-CNN to realize parameter reduction. Besides, Zhu et al. [16] propose a spectral–spatial-dependent global learning framework based on the global convolutional long short-term memory and global joint attention mechanism for insufficient and imbalanced HSIC.

RNNs are prevalent in the fields of machine translation and speech recognition, the key of which is to construct sequential

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data. For HSI data, an intuitive construction method is to consider all spectral bands of a single pixel as a sequence. However, there are generally hundreds of bands in HSI, which may result in gradient vanishing or explosion. LSTM [12] strives to alleviate this problem via the parameters in its forget gate, which are relevant to the degree of gradient vanishing. GCNs obtain satisfactory results of ground object classification by mining both short-range and long-range spatial relations between samples, while CNNs only take short-range spatial relations into account. MGLN [13] learns the graph structural information at both local and global levels in an end-to-end fashion for HSIC, while MDGCN [14] comprehensively extracts the multiscale information inherited by HSIs through dynamic and multiscale GCN. Su et al. [17] integrate a cuckoo search strategy into the trained GCN to adaptively refine the graph structure. Su et al. [18] design a kernel-based iterative filter to avoid the oversmoothing of local features, which often occurs in graph learning. Furthermore, superpixel-based graph attention networks (GATs) and pixel-based CNN have been proved to be complementary by [19]. In recent years, the attention mechanism has become a hot topic. When introduced into deep networks, it simplifies the network by simulating the way that the human brain processes vision. SSFTT [20], FPGA [21], and ViT [22] are all high-performance networks that incorporate the attention mechanism. In order to guarantee that the training and test sets of HSIs follow the same feature distribution, a weakly pseudosupervised decorrelated subdomain adaptation framework is proposed by [23]. Similarly, several studies, such as [24] and [25], also focus on cross-domain or cross-scene HSIC.

It is easy to notice that all the above methods are supervised methods, as they exploit annotation information. Undeniably, those supervised methods are capable of extracting more discriminative features with the involvement of annotations. However, in practical applications, annotations often require expensive labor costs and time consumption. Paradoxically, deep networks do require a large amount of labeled data. The phenomenon named data-hungry makes supervised methods prone to overfitting and generalization problems. A strong candidate is the emerging self-supervised learning (SSL) method. The SSL method does not involve any manual label, which is supposed to be regarded as a branch of unsupervised learning but actually falls into the category of the supervised paradigm [26]. In principle, it aims to learn transferable knowledge from a large amount of unlabeled data through well-designed pretext tasks and then generalize the learned knowledge to downstream tasks with specific supervision signals [27].

The generative-based SSL method has the powerful ability to restore the distribution of raw data without any assumption for downstream tasks. It aims to make the input and output as similar as possible. Unfortunately, generative-based SSL networks are particularly sensitive to raw data, suffering from the overly conservative data distribution. Compared with the generative-based SSL method, the contrastive-based SSL method is often more prominent in performance. The reason may be that it discards the decoder, making it generally lightweighted. The end-to-end contrastive-based SSL method is the most prevalent paradigm, which minimizes the distance between positive pairs and maximizes the distance between negative pairs during training. Therefore, it extremely relies on data augmentations and a terrific amount of tricky negative samples. According to the learning objectives, the generative-based SSL method pays more attention to pixel-level features, while the contrastive-based SSL method cares more about sample-level features. This is natural and reasonable to remind us of such a question: does hyperspectral image classification (HSIC) benefit from the effective combination of the above two SSL methods? We think it does, so we come up with the MECL and prove the benefits of the combination in the later statement.

Overall, the main contributions of this article are summarized as follows.

1) In order to integrate the merits of generative-based SSL methods and contrastive-based SSL methods, we propose an efficient SSL method for HSIC, that is, MECL. Essentially, MECL is still a contrastive-learning (CL) paradigm, but it incorporates the idea of masking and predicting of the generative-based SSL method.
2) In the light of the characteristics of HSI data, we design a spatial–spectral multimasking mechanism. Both pure spectral masking and pure spatial masking will make the reconstruction task too difficult for the network. Hence, the spatial–spectral multimasking mechanism is required to simplify the reconstruction tasks while carrying out efficient denoising.
3) To prevent our MECL from collapsing, we put forward two complementary strategies, namely, target precising and mean-entropy maximization. With the help of these two simple strategies, MECL exhibits outstanding performance.

II. Related Work

A. Generative-Based SSL Methods for HSIC

The autoencoder and the generative adversarial network (GAN) are two typical algorithms of generative-based SSL methods. The autoencoder is commonly used in conjunction with other classification networks in SSL methods for HSIC [28]. For example, Miao et al. [29] integrate the autoencoder into the Siamese network to obtain discriminant features from limited labeled samples. Jia et al. [30] rectify unsupervised generative features learned by 3-D encoders indirectly with limited labeled samples in the Siamese network. Ibañez et al. [31] embed the masked autoencoder into the spectral–spatial Transformer for the purpose of utilizing its denoising ability to gain more powerful representations. Compared with AEs, GANs occupy a core position in the hyperspectral field. Zhu et al. [32] first explore the usefulness and effectiveness of GAN for HSIC. Bai et al. [33] design multiple metrics to jointly optimize the generator for high-quality generated samples. Wang et al. [34] encourage the dual-channel network generated by coupling the capsule network with GANs to further explore the spectral–spatial information of HSI data. In addition, the GAN is flexibly excogitated to address the existing issue of severe category

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imbalance in HSI datasets. Roy et al. [35] implement minority oversampling via the GAN, intending to enforce their network to automatically create more generated samples for the few-sample classes. Feng et al. [36] design a discriminator with multiple specific branches and a general branch to weaken the generation preference for the large-size classes. Qin et al. [37] add two distance constraints to the GAN architecture to generate enough samples for each class and improve the quality of the generated samples.

B. Contrastive-Based SSL Methods for HSIC

Most end-to-end contrastive-based SSL methods in hyperspectral field employ CNN architecture for feature extraction. Zhu et al. [38] exploit lightweight convolution units with asymmetric multiscale and multishape receptive field to capture multilevel features of the same pixel. A cross-domain CNN-based CL algorithm aiming at maximizing the similarity between the spatial representation and the spectral representation of the same sample is proposed by [39] for the first time, which strongly confirmed that the contents useful for HSIC are invariant in the spectral and spatial domains. Hang et al. [40] conduct cross-modality CNN-based CL based on the relationship between hyperspectral data and LiDAR data. Inspired by the landmark work SimCLR [41], the deep residual network has also been applied to the end-to-end contrastive-based method in hyperspectral field. Hou et al. [42] utilize random Gaussian noise to generate positive and negative pairs and then use ResNet50 to extract abstract HSI representations. Zhao et al. [43] put forward a series of augmentations for HSI and adopt ResNet18 as the encoder.

In addition, a lot of end-to-end CL methods based on other architectures have also been introduced to HSIC. A hierarchical CL method based on the Transformer is proposed by [44] to promote the network and learn the multiscale semantic representations of HSI adaptively. Zhou et al. [45] combine the simplified vision Transformer with CL to alleviate the oversmoothing problem and increase the differences of interclass features. Yu et al. [46] introduce GCN into the CL framework for the first time and come up with several improved graph augmentation operations. However, the practice of maximizing the distances between negative pairs in the traditional CL framework probably leads to the separation of intraclass samples in the feature space. Consequently, Lu et al. [47] promote their network to learn HSI representations with higher intraclass compactness and interclass discreteness by means of distance assessments. Jiang et al. [48] employ graph structure learning to generate global views and use absolute positional encoding to assist in the generation of local views. Besides, Liu et al. [49] first integrate CL and prototypical-based few-shot learning (FSL) into an end-to-end network for HSIC.

C. Hybrid SSL Methods for HSIC

With an eye on the state-of-the-art research, hybrid SSL methods have received substantial interest in the hyperspectral field. Yang et al. [50] utilize image rotation, image scale, and spectral order to generate auxiliary labels and combine auxiliary learning and CL tasks ingeniously. Li et al. [51] incorporate random occlusion into the end-to-end BYOL to address the strong dependence of CL on numerous negative samples and enhance the robustness of hyperspectral classification algorithms. But, they regard masking as a new way of data augmentations, not realizing that the training process of CL with masking is actually an implicit feature reconstruction. In an effort to improve the performance of downstream classification tasks, Huang et al. [52] construct an end-to-end CL network on the basis of the spectral–spatial masked Transformer (SS-MTr). Their network predicts missing pixels with the help of SS-MTr to mine the local spatial relationship in HSI data, while engaging end-to-end CL for more discriminative information. However, they only adopt the common random spatial masking, without putting forward an effective masking strategy according to the characteristics of hyperspectral data. Cao et al. [53] combine the prototypical CL method and the classic representation learning methods, including the AE and the GAN, to design an unsupervised feature learning network for hyperspectral classification. Their work is most similar to ours, but we adopt the strategy of masked AEs (MAEs) indirectly. In general, our method MECL does not predict missing values at the input level, but instead performs reconstruction implicitly by matching masked online views with the unmasked target view.

III. Methodology

In this chapter, we first summarize the input views and the overall framework of MECL. Then, we introduce the main modules of MECL in detail. Finally, we clarify our learning objective or loss function in terms of its utility.

A. Overall Framework

1) Input Views: The original HSI cube is denoted as $\mathbf{I} \in \mathcal{R}^{M\times N\times C}$, where M, N, and C are the width and height of the HSI cube and the number of spectral bands, respectively. First of all, a principal component analysis (PCA) is performed on $\mathbf{I}$ to greatly reduce spectral redundancy in the original HSI cube as well as accelerate network training. The image cube after PCA is $\mathbf{X} \in \mathcal{R}^{M\times N\times B}$, where B is the number of spectral bands after PCA. Then, we extract spatial–spectral 3-D patches of $S \times S \times B$ size for each pixel from $\mathbf{X}$, where S is the height/width of the corresponding patch and can only be a multiple of 3. These patches serve as the input views of MECL, the labels of which are decided by the labels of their central pixels. For the illustrative purpose, we describe the ith patch as $\mathbf{x}_i$.

2) Our Work (MECL): MECL consists of two networks, the online network and the target network (Fig. 1). The pseudo code of MECL is provided in Algorithm 1. Specially, the two networks involve three main modules: spatial–spectral multimasking module (2S3M), shared feature extraction module (SFEM), and prototype-based CL module (PCLM). The data flow of MECL will be briefly introduced below. First of all, MECL performs data augmentations on the input view $\mathbf{x}_i$ by using $m+1$ groups of random augmentations $\mathbf{T}_{1\sim m+1}$, in order to generate m online views $\mathbf{x}_{i,1\sim m}$ for the input of the online network and one target view $\mathbf{x}_i^+$ for the input of the target network, where $m > 1$ is required.

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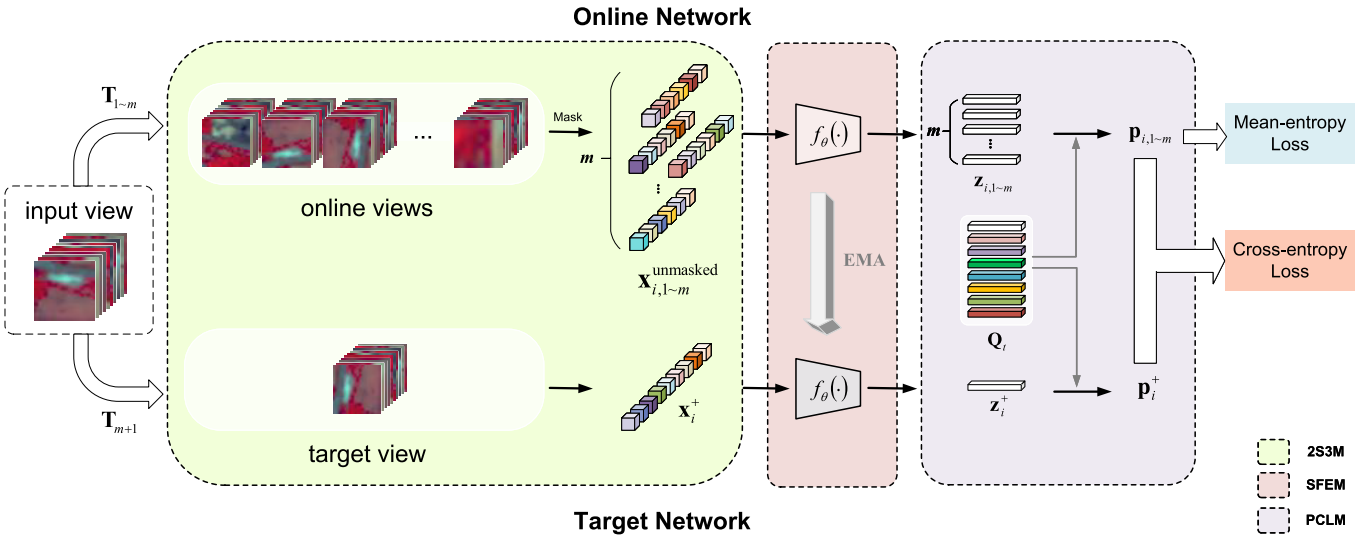

Fig. 1. Framework of MECL. MECL consists of two networks, namely, the online network and the target network. These networks involve three main modules, namely 2S3M, SFEM, and PCLM. First, MECL performs data augmentations on the input view $\mathbf{x}_i$ to generate m online views $\mathbf{x}_{i,1\sim m}$. After data augmentations, 2S3M performs the spatial–spectral multimasking operation on $\mathbf{x}_{i,1\sim m}$ of the online network to obtain $\mathbf{x}_{i,1\sim m}^{\text{unmasked}}$. Next, MECl feeds both $\mathbf{x}_{i,1\sim m}^{\text{unmasked}}$ and $\mathbf{x}_i^+$ to SFEM for feature extraction to obtain online representations $\mathbf{z}_{i,1\sim m}$ and target representation $\mathbf{z}_i^+$. Then, PCLM updates the prototype set $\mathbf{Q}_t$ via the representations. Finally, MECL utilizes the cross-entropy loss and the mean-entropy loss to update the parameters in these two networks.

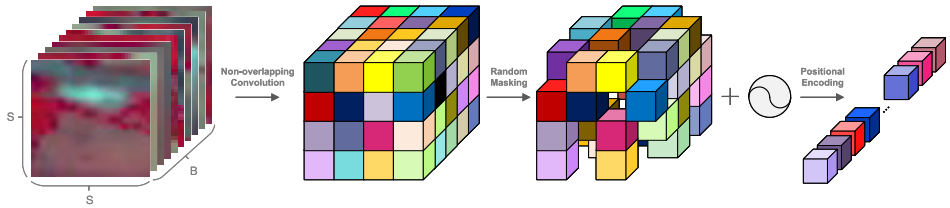

Fig. 2. Masking strategy in spatial–spectral multimasking module.

After data augmentations, 2S3M performs the spatial–spectral multimasking operation on $\mathbf{x}_{i,1\sim m}$ of the online network to obtain $\mathbf{x}_{i,1\sim m}^{\text{unmasked}}$. Notably, 2S3M does not mask the target view $\mathbf{x}_i^+$. Next, MECl feeds both $\mathbf{x}_{i,1\sim m}^{\text{unmasked}}$ and $\mathbf{x}_i^+$ to SFEM for feature extraction to obtain online representations $\mathbf{z}_{i,1\sim m}$ and target representation $\mathbf{z}_i^+$. Then, PCLM updates the prototype set $\mathbf{Q}_t$ via the representations and calculates the similarities $\mathbf{p}_{i,1\sim m}$ and $\mathbf{p}_i^+$ between the prototype set and $\mathbf{z}_{i,1\sim m}$ and $\mathbf{z}_i^+$, respectively. Finally, MECL utilizes the well-designed loss function to update the parameters in the two networks.

B. Data Augmentations

According to the characteristics of HSI, we choose two frequently used and effective data augmentations: random cropping and random flipping. Composing augmentation pairs of them benefits more in our work, since it can further improve the performance. In this part, multiple online views are generated to fully exploit the distinguishing features in the original input of MECL and increase the diversity of instances.

C. Spatial–Spectral Multimasking Module

The spatial–spectral multimasking module is divided into three important steps, namely, nonoverlapping convolution, masking, and position encoding. Fig. 2 shows the three steps performed on one of the online views $\mathbf{x}_{i,j}$, where $j \in [1, m]$. Unlike traditional masking operations, 2S3M masks the abstract features extracted by 3D-CNN, thus encouraging the network to learn more discriminative spatial–spectral information through implicit reconstruction.

1) Nonoverlapping Convolution: In this part, the online views $\mathbf{x}_{i,1\sim m}$ are convolved with the 3D-CNN kernel for feature extraction. Seriously, it is essential to carry out nonoverlapping convolution, because the subsequent masking object is a whole small cube, and we do not want to disturb SFEM manually too much so as to prompt it to learn representative knowledge autonomously. The nonoverlapping convolution process is equivalent to first regarding the online views $\mathbf{x}_{i,1\sim m}$ as the collections $\{\mathbf{c}_g\}_{i,1\sim m}$ composed of many $3 \times 3 \times 3$ small cubes $\mathbf{c}_g$, followed by extracting the features of each small cube $\mathbf{c}_g$ via the $3 \times 3 \times 3$ kernel to obtain embedding sets $\{\mathbf{e}_g\}_{i,1\sim m}$. Obviously, the value of g ranges from 1 to $(S \times S \times B)/27$. To wrap up, this simple convolution operation eliminates redundant and highly relevant information and induces MECL to focus on more important information.

2) Masking: We employ random masking on the small cubes $\{\mathbf{e}_g\}_{i,1\sim m}$. In this process, some small cubes with discontinuous positions are randomly removed by 2S3M to

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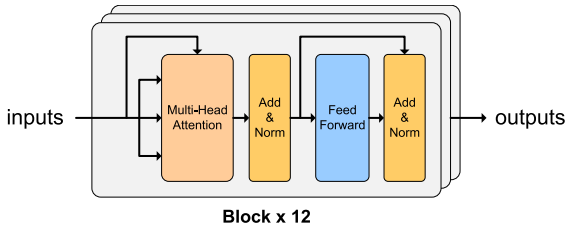

Fig. 3. Structure of the encoder.

obtain $\{\mathbf{e}_g\}_{i,1\sim m}^{\text{unmasked}}$. The masked cubes are no longer propagated forward, while the unmasked cubes are thrown into SFEM, which greatly shortens the training time and reduces the demand for hardware.

3) Positional Encoding: In order to make full use of spatial information in HSI, the relative position information of small cubes must be injected. For this reason, 2S3M adds positional encodings to the embedding sets $\{\mathbf{e}_g\}_{i,1\sim m}^{\text{unmasked}}$ for generating $\mathbf{x}_{i,1\sim m}^{\text{unmasked}}$. The sine and cosine functions of different frequencies in the prevalent Transformer [54] are adopted by our work to complete positional encoding.

D. Shared Feature Extraction Module

The SFEM is made up of two encoders from the online network and the target network. The structure of such two encoders is the same, which is composed of 12 standard blocks of the Transformer [54] (Fig. 3). The parameterized encoder is defined as $f_\theta(\cdot)$, and the parameters of the online encoder and the target encoder are denoted as θ_{online} and θ_{target}, respectively. During training, θ_{online} is updated according to the loss, while θ_{target} is updated by the exponential moving average (EMA) technique, which can be formulated as follows:

$$\theta_{\text{target}} = \alpha\theta_{\text{target}} + (1-\alpha)\theta_{\text{online}} \tag{1}$$

where α is the balance coefficient, generally taking values in the range [0.9, 0.9999]. What matters is that, the EMA controls the differences in feature sequences extracted from the online network and the target network to a certain extent.

E. Prototype-Based CL Module

For the purpose of solving the existing problems of the popular end-to-end CL, we put forward the prototype-based CL module. As is known to all, the classical end-to-end CL always draws positive pairs closer and pushes negative pairs apart. In contrast, PCLM no longer requires negative samples, which means it can get rid of the dependence on a host of negative samples. PCLM calculates the similarities between multiple online views and a single target view and a fixed number of prototypes, respectively, and then punishes the similarity differences caused by encoders. The implementation of PCLM includes two steps, namely, prototype set updating and similarity calculation.

1) Prototype Set Updating: As shown in formula (2), the prototype set of the current batch $\mathbf{Q}_t \in \mathcal{R}^{k\times d}$ is generated by clustering via K-means in PCLM, where k is the number of prototypes and d is the dimension of each prototype. In detail, PCLM feeds the representation $\mathbf{z}_t$, jointly obtained by online network and target network, and the prototype set of the last batch $\mathbf{Q}_{t-1}$ into K-means to generate $\mathbf{Q}_t$, i.e.,

$$\mathbf{Q}_t = \text{K}-\text{means}(\mathbf{Q}_{t-1}, \mathbf{z}_t) \tag{2}$$

where $\mathbf{z}_t$ with a length of $b\times(m+1)$ includes all $\mathbf{z}_{i,1\sim m}$ and $\mathbf{z}_i^+$ in the current batch and b is the size of the current batch. As the number of training batches increases, PCLM enables the prototype set $\mathbf{Q}_t$ to characterize the whole data gradually.

2) Similarity Calculation: PCLM calculates the dot product of a single representation $\mathbf{z}_i$ with the prototype set $\mathbf{Q}_t$, namely, cosine similarity, to measure the matching degree between $\mathbf{z}_i$ and all prototypes. We desire to mitigate the differences between the matching degrees of online representations and the target representation. The formula for similarity calculation can be written as follows:

$$\mathbf{p}_i = \text{softmax}\left(\frac{\mathbf{Q}_t \cdot \mathbf{z}_i^{\text{T}}}{\tau}\right) \tag{3}$$

where τ is a temperature, ranging from 0 to 1. We stipulate that $\mathbf{p}_{i,1\sim m}$ corresponds to τ_{online}, while $\mathbf{p}_i^+$ corresponds to τ_{target}. To explicitly ensure a more precise target representation produced by MECL and implicitly guide MECL to generate online representations with low entropy, the constraint $\tau_{\text{target}} < \tau_{\text{online}}$ has been always maintained, called "target precising" in this article. In addition, we have found that combining "target precising" with our loss function can effectively avoid MECL from outputting constant.

Algorithm 1 Pseudocode of MECL

```
imgs = patch(PCA(data))
loader = dataloader(dataset(imgs, labels), batch_size)
for batch in loader do
  for i in batch do
    # Part1: 2S3M
    x_i^+, x_{i,1~m} = T_{1~m+1}(x_i) # Data Augmentations
    x_{i,1~m}^unmasked = Conv&Mask(x_{i,1~m})
    # Part2: SFEM
    z_i^+ = f_{θ_target}(x_i^+)
    z_{i,1~m} = f_{θ_online}(x_{i,1~m}^unmasked)
    # Part3: PCLM
    Q_t = K − means(Q_{t−1}, z_t) # Prototype Set Updating
    p_i = softmax(Q_t·z_i^T / τ_target)
    p_{i,1~m} = softmax(Q_t·z_{i,1~m}^T / τ_online)
  end for
  # cross-entropy loss
  closs = 1/bm Σ_{i=1}^{b} Σ_{j=1}^{m} H(p_i^+, p_{i,j})
  # mean-entropy loss
  p̄ = 1/bm Σ_{i=1}^{b} Σ_{j=1}^{m} p_{i,j}
  rloss = H(p̄)
  loss = closs − λ * rloss
  loss.backward()
  optimizer.step()
end for
```

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TABLE I
CATEGORY INFORMATION OF THE INDIAN PINES DATASET

No.	Class	Train	Test
1	Alfalfa	20	26
2	Corn-notill	20	1408
3	Corn-mintill	20	810
4	Corn	20	217
5	Grass-pasture	20	463
6	Grass-trees	20	710
7	Grass-pasture-mowed	20	8
8	Hay-windrowed	20	458
9	Oats	15	5
10	Soybean-notill	20	952
11	Soybean-mintill	20	2435
12	Soybean-clean	20	573
13	Wheat	20	185
14	Woods	20	1245
15	Bldg-gress-trees	20	366
16	Stone-Steel-Towers	20	73

TABLE II
CATEGORY INFORMATION OF THE WHU-Hi-HanChuan DATASET

No.	Class	Train	Test
1	Strawberry	20	44715
2	Cowpea	20	22733
3	Soybean	20	10267
4	Sorghum	20	5333
5	Water-spinach	20	1180
6	Watermelon	20	4513
7	Greens	20	5883
8	Trees	20	17958
9	Grass	20	9449
10	Red-roof	20	10496
11	Gray-roof	20	16891
12	Plastic	20	3659
13	Bare-soil	20	9096
14	Road	20	18540
15	Bright-object	20	1116
16	Water	20	75381

F. Learning Objective

On the one hand, the cross-entropy loss is $\mathrm{H}(\mathbf{p}_i^+, \mathbf{p}_{i,j})$, where $j \in [1, m]$ is adopted to penalize when the online representation $\mathbf{p}_{i,j}$ differs from the target representation $\mathbf{p}_i^+$.

On the other hand, in order to further prevent the network from collapsing, we also introduce a mean-entropy loss into our loss function. It can force representations to match more different prototypes, thereby bringing more interesting challenges to the MECL. The average representation across all the online views $\bar{\mathbf{p}}$ is expressed as follows:

$$\bar{\mathbf{p}} = \frac{1}{bm} \sum_{i=1}^{b} \sum_{j=1}^{m} \mathbf{p}_{i,j}. \tag{4}$$

Therefore, as shown in (5), our loss function consists of the cross-entropy loss term (the former) and the mean-entropy loss term (the latter), where $\lambda > 0$ is the balance coefficient

$$\mathcal{L} = \frac{1}{bm} \sum_{i=1}^{b} \sum_{j=1}^{m} \mathrm{H}\left(\mathbf{p}_i^+, \mathbf{p}_{i,j}\right) - \lambda \mathrm{H}\left(\bar{\mathbf{p}}\right). \tag{5}$$

IV. EXPERIMENTS

A. Data Description

We evaluate the proposed MECL against three widely used hyperspectral datasets: Indian Pines, WHU-Hi-HanChuan, and Houston University 2013.

1) Indian Pines: The Indian Pines dataset was acquired by the AVIRIS sensor over the agricultural Indian Pine test site in northwestern Indiana, USA. The scene includes 145 × 145 pixels and is composed of 16 main land-cover categories, with 10 249 samples available for classification. It has 224 spectral reflectance bands, but not all of them are available. Before the experiment, we have retained 200 available bands by removing 20 water absorption bands. The class name and the number of samples for training and testing in the dataset are listed in Table I.

2) WHU-Hi-HanChuan: The WHU-Hi-HanChuan dataset was captured by the Headwall Nano-Hyperspec imaging sensor equipped on a Leica Aibot X6 UAV V1 platform in Hanchuan, Hubei, China. The scene consists of 274 bands, covering the spectral range from 400 to 1000 nm. It is 1217 × 303 in size and contains 16 main land-cover categories, with up to 257 530 pixels available for classification. The detailed category information of the WHU-Hi-HanChuan dataset is provided in Table II.

3) Houston University 2013: The Houston University 2013 dataset was published as part of the 2013 IEEE Geoscience and Remote Sensing Society (GRSS) data fusion contest. The scene contains 349 × 1905 pixels, with 144 spectral bands. The Houston University 2013 dataset contains 15 classes and 15 029 labeled pixels available for classification. The amounts of training samples and testing samples of various classes are summarized in Table III.

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TABLE III
CATEGORY INFORMATION OF THE HOUSTON UNIVERSITY 2013 DATASET

No.	Class	Train	Test
1	Grass Healthy	20	1231
2	Grass Stressed	20	1234
3	Grass Synthetic	20	677
4	Tree	20	1224
5	Soil	20	1222
6	Water	20	305
7	Residential	20	1248
8	Commercial	20	1224
9	Road	20	1232
10	Highway	20	1207
11	Railway	20	1215
12	Parking Lot 1	20	1213
13	Parking Lot 2	20	449
14	Tennis Court	20	408
15	Running Track	20	640

B. Experimental Setup

MECL falls into the category of the SSL paradigm, consequently including two stages: pretraining and fine-tuning. During pretraining, no annotations are used, and 50% pixels are selected. During fine-tuning, 20 labeled pixels are randomly taken from each category for training, and the remaining pixels are applied for testing. If the total number of a category is less than 20, take 15.

MECL consists of three main modules, that is, 2S3M, SFEM, and PCLM. In 2S3M, the masking ratios of the Indian Pines dataset, the WHU-Hi-HanChuan dataset, and the Houston University 2013 dataset are 0.3, 0.75, and 0.85, respectively. In SFEM, the balance coefficient α of the EMA technique is set to 0.9. In PCLM, the dimension of a single representation $\mathbf{z}_i$ is set to 256, and the number of all prototypes in the prototype set is fixed to 1024. What is more, Adam optimizer is employed to optimize the network, the learning rate is initialized to be 0.001, the balance coefficient λ of the loss function is fixed to 1, and the total number of epochs is set to 200.

To testify the effectiveness of MECL, we compare it to seven representative methods: a classical classification method, i.e., SVM-RBF; two CNN-based methods, i.e., HybridSN [15] and SSRN [11]; three attention-based methods, i.e., SSFTT [20], FPGA [21], and ViT [22]; and an SSL-based method, i.e., SimSiam [55]. Three assessment criteria, namely, overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa), are chosen to evaluate the performance of these methods. In brief, the higher the values of them, the better the performance. Moreover, all experiments are conducted on a high-performance computer equipped with GeForce RTX 3090 GPU with 24G memory and Intel[1] Xeon[1] Silver 4214 CPU, so as to make our experimental results more convincing.

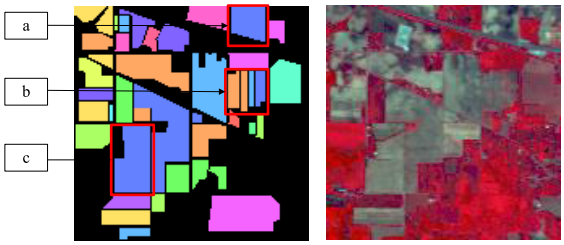

Fig. 4. Indian Pines. (a) Ground-truth map. (b) False-color image.

C. Classification Results

In an attempt to validate the classification performance of our MECL quantitatively and qualitatively, we conduct comparative experiments of MECL and the aforementioned seven baselines on three real HSI datasets. The classification results of different methods on each dataset are listed in Tables IV–VI. The values in three tables are the means obtained through ten repeated experiments, to ensure the persuasiveness of our experiments. Besides, for the convenience of analysis, we marked areas with significant differences in classification maps via red rectangular frames in the three ground-truth maps of Figs. 4–6. Furthermore, the classification maps of each method on three datasets are shown in Figs. 7–9, to visually and intuitively display the classification results of different methods.

1) Results on Indian Pines: The SVM-RBF only uses spectral information, which leads to the worst performance. Although HybridSN combines 2-D convolution and 3-D convolution in a sequential manner, it cannot effectively filter out the noise in Indian Pines dataset. SSRN, benefiting from the fusion of spatial branch and spectral branch, still cannot gain enough discriminant information when there are only 20 labeled samples in each class. Both SSFTT and FPGA achieve good classification results, while ViT behaves best in supervised baselines. However, compared with MECL, ViT still lags behind 8.30% in OA. The performance of SimSiam is second only to MECL, with an OA of 87.53%. Moreover, MECL wins the best classification accuracies in 12 out of 16 categories in the Indian Pines dataset.

It can be intuitively observed in Fig. 7 that the noise in pure spectral-based SVM-RBF is quite severe. Except for SimSiam and MECL, there has been a common phenomenon that large areas of ground objects are misjudged in region *a* of other methods. Perhaps, these methods do not fully utilize the spatial information of adjacent pixels. Only a small number of pixels in region *a* are misclassified by SimSiam,

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TABLE IV
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON INDIAN PINES

No.	SVM-RBF	HybridSN	SSRN	SSFTT	FPGA	ViT	SimSiam	MECL
1	47.11	61.50	**100.00**	**100.00**	97.44	**100.00**	**100.00**	**100.00**
2	54.82	72.24	63.67	82.17	62.77	66.11	82.95	**92.33**
3	45.14	64.62	66.77	89.48	82.87	72.53	**92.22**	87.65
4	47.14	54.45	95.86	95.12	98.70	**100.00**	99.54	**100.00**
5	73.43	89.03	90.31	**92.44**	90.76	88.02	83.80	83.37
6	88.49	73.15	90.25	99.44	96.68	95.07	99.30	**99.72**
7	40.26	75.96	**100.00**	**100.00**	**100.00**	92.86	**100.00**	**100.00**
8	93.08	78.61	98.03	**100.00**	**100.00**	**100.00**	99.78	**100.00**
9	46.89	26.73	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
10	55.93	71.49	76.30	84.14	82.38	87.24	88.45	**92.02**
11	57.92	81.82	74.63	61.34	70.47	72.64	77.74	**86.69**
12	49.11	73.70	63.78	72.15	87.71	83.84	84.12	**91.97**
13	86.55	75.16	99.51	95.68	**100.00**	**100.00**	**100.00**	**100.00**
14	84.15	84.35	96.98	97.46	98.41	**99.21**	91.41	92.21
15	50.55	71.89	95.28	95.74	99.74	**100.00**	98.91	99.45
16	95.06	51.14	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
OA(%)	63.13	73.17	79.93	82.97	82.95	83.32	87.53	**91.62**
AA(%)	73.38	69.11	88.21	91.57	91.74	91.09	93.64	**95.34**
kappa($\times$100)	58.57	69.97	77.22	80.81	80.72	81.25	85.86	**90.48**

TABLE V
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON WHU-Hi-HanChuan

No.	SVM-RBF	HybridSN	SSRN	SSFTT	FPGA	ViT	SimSiam	MECL
1	74.31	90.09	73.71	89.48	92.71	91.51	**97.31**	96.70
2	44.88	**88.84**	68.60	78.89	63.89	73.50	79.11	86.91
3	43.03	65.94	82.74	88.42	95.84	83.80	**97.70**	96.41
4	66.10	58.91	**98.16**	97.88	93.22	96.23	97.84	97.92
5	18.16	23.47	98.07	98.92	99.24	85.93	**100.00**	**100.00**
6	19.27	46.71	64.40	71.85	92.22	57.97	**92.42**	91.29
7	55.12	66.56	93.64	94.14	**96.25**	94.54	93.91	95.94
8	42.45	68.65	71.92	66.56	81.19	62.33	83.33	**84.58**
9	30.96	68.02	72.50	84.03	**90.59**	75.31	82.80	85.50
10	70.99	75.60	89.64	92.44	**95.44**	94.93	91.48	93.24
11	57.68	85.99	91.46	94.60	86.01	87.44	**99.44**	99.37
12	23.66	39.20	87.78	91.63	**97.11**	94.59	94.12	94.15
13	28.32	51.85	53.69	63.04	61.84	62.07	69.23	**72.81**
14	62.60	71.92	67.93	78.67	87.23	71.97	92.69	**93.93**
15	36.52	33.24	93.27	96.16	83.37	93.37	**99.91**	99.82
16	87.82	96.92	**98.39**	97.82	97.40	97.94	98.01	97.86
OA(%)	62.14	79.14	82.57	87.96	89.10	85.89	92.81	**93.80**
AA(%)	56.30	64.49	81.62	86.53	88.35	82.71	91.83	**92.90**
kappa($\times$100)	56.95	75.79	79.84	85.99	87.30	83.55	91.62	**92.77**

where MECL is able to achieve precise classification. The same issue also occurs in region *c*. Except MECL, other methods are affected by surrounding pixels to varying degrees, which causes wrong classification results. For region *b* with a complex layout, the classification results of each method are not ideal. The most likely reason is that there are two very similar land cover types in region *b*, namely, Soybean-notill and Soybean-mintill. But, on the whole, MECL manifests the best classification performance in the Indian Pines dataset.

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TABLE VI
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON HOUSTON UNIVERSITY 2013

No.	SVM-RBF	HybridSN	SSRN	SSFTT	FPGA	ViT	SimSiam	MECL
1	98.38	98.21	84.96	97.89	**98.96**	97.73	97.36	96.72
2	94.08	95.95	**99.71**	97.97	94.98	91.98	94.66	95.53
3	98.23	99.56	**100.00**	99.70	99.86	99.41	98.42	99.28
4	89.87	98.45	97.52	95.34	98.47	**99.10**	93.89	98.55
5	99.43	**100.00**	96.98	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
6	97.38	98.36	96.46	**100.00**	93.54	95.41	98.15	98.15
7	71.71	92.95	91.38	85.58	92.59	87.58	88.56	**93.53**
8	58.58	79.33	**99.17**	88.32	81.59	86.76	79.74	85.45
9	77.11	79.95	90.71	86.61	91.69	87.58	92.49	**96.41**
10	86.91	**97.27**	87.21	87.74	93.48	95.11	86.63	94.78
11	76.95	87.90	93.12	93.58	80.16	87.33	89.39	**99.60**
12	73.70	86.07	**94.25**	90.11	71.61	78.65	59.21	81.59
13	44.10	96.88	83.30	95.32	93.39	97.77	95.52	**97.87**
14	97.06	**100.00**	89.08	**100.00**	97.66	**100.00**	99.07	**100.00**
15	98.75	**100.00**	99.69	99.53	**100.00**	**100.00**	**100.00**	**100.00**
OA(%)	83.61	92.87	93.24	93.43	91.62	92.48	89.97	**95.08**
AA(%)	84.15	94.06	93.57	94.51	92.53	93.63	91.54	**95.83**
kappa($\times$100)	82.29	92.29	92.69	92.89	90.94	91.87	89.16	**94.68**

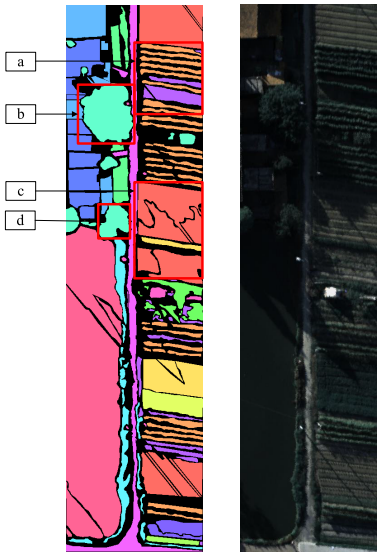

Fig. 5. WHU-Hi-HanChuan. (a) Ground-truth map. (b) False-color image.

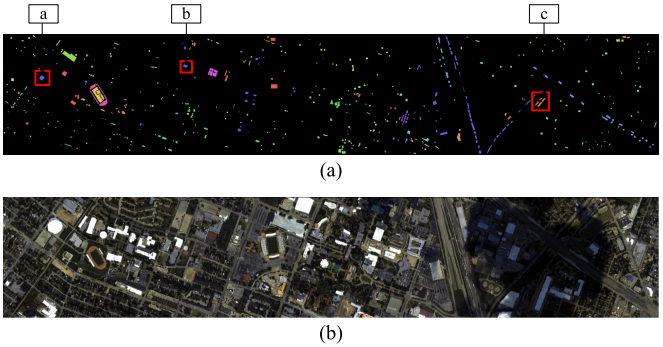

Fig. 6. Houston University 2013. (a) Ground-truth map. (b) False-color image.

2) Results on WHU-Hi-HanChuan: The performance differences of the listed methods on the WHU-Hi-HanChuan dataset are conspicuous, which can be observed in Table V. Specifically, SVM-RBF still obtains the worst performance. HybridSN performs spatial–spectral fusion in series, achieving an OA improvement of 18.84% compared with SVM-RBF. SSRN employs a parallel method to extract spatial–spectral information, achieving an OA improvement of 4.05% compared with HybridSN. This indicates that the method obtaining joint spatial–spectral information in parallel is more effective on large-scale datasets with certain classification difficulties. Among attention-based baselines, FPGA performs best through attaching importance to long-distance spatial dependence. Here, FPGA further verifies its natural advantages in ultralarge images. In addition, only the indicators of MECL and SimSiam conforming to the contrastive-based SSL paradigm exceed 90%. Although MECL only achieves the best accuracy in 4 out of 16 categories in the WHU-Hi-HanChuan dataset, its three criteria are the highest, which proves that its classification performance in each category is stable.

Visually, there are great differences in the classification maps within the given four regions in Fig. 5. Without the involvement of spatial information, SVM-RBF still performs the worst. The performances of HybridSN and SSRN are not ideal, because the ground distribution of the dataset is relatively complicated. Overall, the attention-based methods significantly outperform the CNN-based methods on complex

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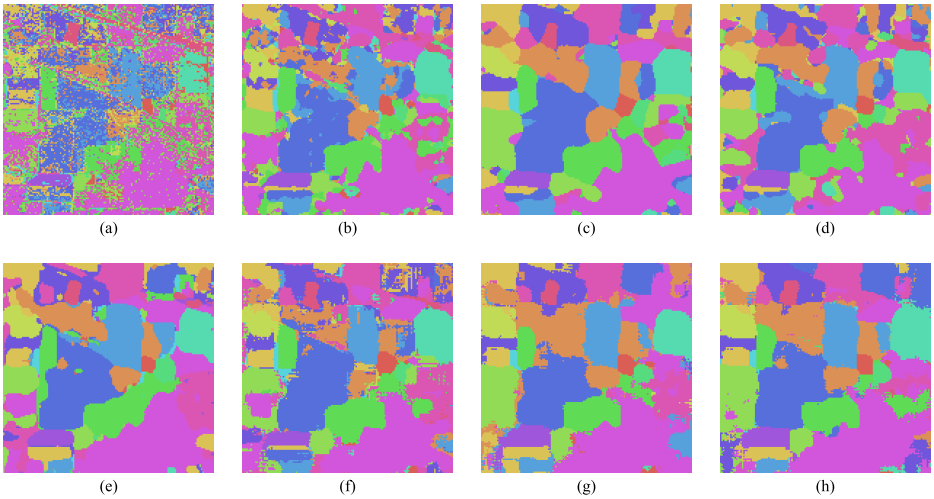

Fig. 7. Classification maps of different methods on the Indian Pines dataset. (a) SVM-RBF. (b) HybridSN. (c) SSRN. (d) SSFTT. (e) FPGA. (f) ViT. (g) SimSiam. (h) MECL.

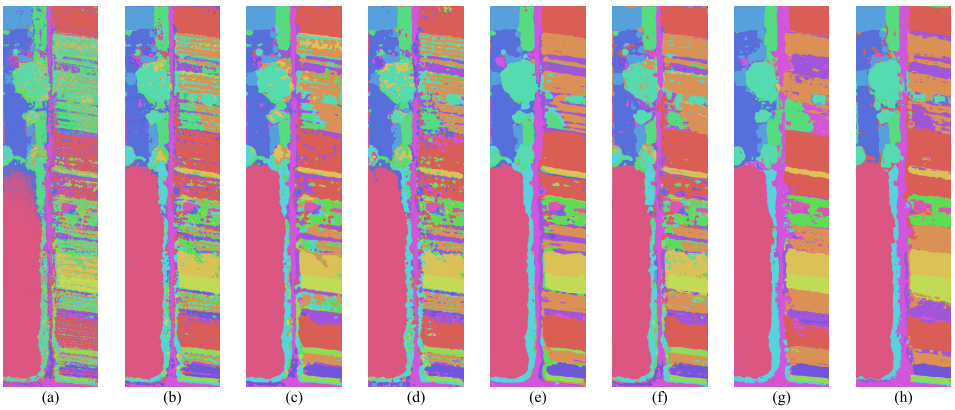

Fig. 8. Classification maps of different methods on the WHU-Hi-HanChuan dataset. (a) SVM-RBF. (b) HybridSN. (c) SSRN. (d) SSFTT. (e) FPGA. (f) ViT. (g) SimSiam. (h) MECL.

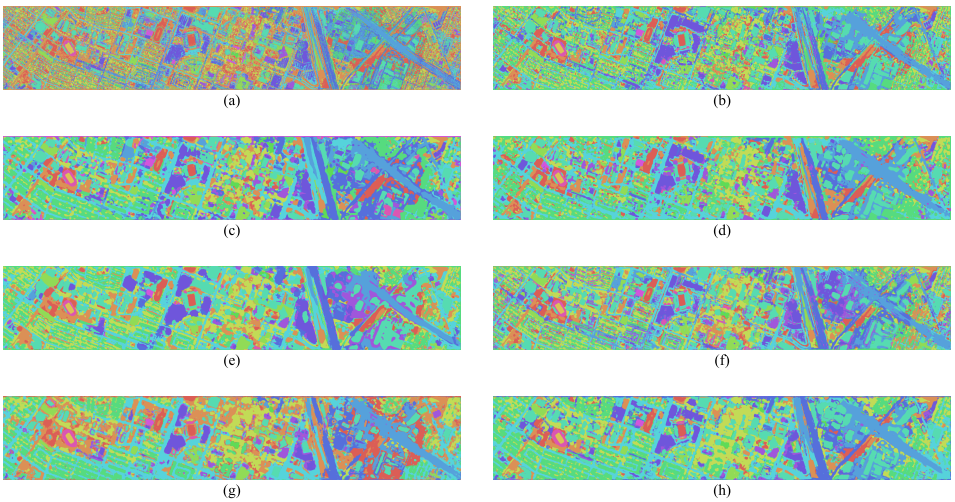

Fig. 9. Classification maps of different methods on the Houston University 2013 dataset. (a) SVM-RBF. (b) HybridSN. (c) SSRN. (d) SSFTT. (e) FPGA. (f) ViT. (g) SimSiam. (h) MECL.

classification tasks. As expected, FPGA is good at classifying the large area of the same category, but it is easily affected by neighborhood pixels in small region d. The classification map of contrastive-based SimSiam is more accurate than

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TABLE VII
RESULTS OF THE EFFECTIVENESS ANALYSIS OF PRETRAINING ON THREE DATASETS

Data Set	Criterion	Without Pre-training	With Pre-training	Improvement
Indian Pines	OA(%)	74.21	**91.62**	**17.41**
	AA(%)	84.12	**95.34**	**11.22**
	Kappa×100	70.98	**90.48**	**19.50**
WHU-Hi-HanChuan	OA(%)	85.89	**93.80**	7.91
	AA(%)	82.71	**92.90**	10.19
	Kappa×100	83.55	**92.77**	9.21
Houston University 2013	OA(%)	92.09	**95.08**	2.99
	AA(%)	93.47	**95.83**	2.36
	Kappa×100	91.45	**94.68**	3.23

MECL in region *c*, but it is not as good as MECL in region *a*.

3) Results on Houston University 2013: Except SVM-RBF and SimSiam, the OAs of all methods exceed 90% (Table VI). MECL outperforms all other algorithms and has opened a certain gap. SimSiam is an SSL algorithm. Its poor performance shows that its pretext task cannot help model effectively learn the potential knowledge of Houston University 2013 dataset. Among the three algorithms based on self-attention mechanism, SSFTT performs best. The reason may be that SSFTT makes full use of the advantages of the combination of CNN and Transformer. The classification results of SSRN and SSFTT are similar, which may be due to their feature learning from spatial and spectral dimensions.

The classification results of each method are accurate in most regions (Fig. 9), with three regions *a*, *b*, and *c* shown in Fig. 6, presenting significant differences in classification. SVM-RBF, HybridSN, SSRN, and ViT are almost completely wrong in classifying the pixels of region *b*. These methods misclassify Parking Lot 1 as either Highway or Soil. In the same case, FPGA and SimSiam behave similarly in region *a*. Specifically, FPGA misclassifies Parking Lot 1 as Highway, and SimSiam misclassifies Parking Lot 1 as Commercial. MECL only misclassifies some pixels on the edge of region *a*. A possible reason is the high spectral similarity between these land covers, which makes it difficult for these methods to perfectly distinguish them. When it comes to region *c*, SSRN exhibits significant inaccuracies in pixel classification. SimSiam also misclassifies a portion of pixels there. However, ViT and MECL only classify a small number of pixels incorrectly. These algorithms confuse Healthy grass with Stressed grass, as both belong to the category of grass.

D. Effectiveness Analysis of Pretraining

In this section, two training methods, namely, with pretraining and without pretraining, are compared to verify the effectiveness of pretraining on MECL. Through Table VII, we notice that the classification results with pretraining are much better than those without pretraining.

On the Indian Pines dataset, OA, AA, and Kappa have increased by 17.41%, 11.22%, and 19.50%, respectively, after adding pretraining. Compared with the other two datasets, the improvement of the challenging Indian Pines dataset is the most significant, which shows that the pretraining of MECL is competent to fully suppress noise and explore more meaningful semantic information. On the WHU-Hi-Hanchuan dataset, OA, AA, and Kappa have also improved obviously after pretraining. Although the improvement brought by the pretraining on the Houston University 2013 dataset is not as good as the previous two datasets, each indicator has achieved an increase of at least 2%.

Overall, the outstanding improvements of the three criteria on each dataset demonstrate the usefulness and effectiveness of MECL.

E. Ablation Study

MECL consists of three important modules, among which SFEM containing encoders must exist in the network. Therefore, we only carry out ablation study for the remaining two modules. In this section, we set up three sets of experiments by keeping only masking mechanism (2S3M + SFEM), only contrastive-learning mechanism (PCLM + SFEM), and both (MECL). The above three experiments correspond to the generative-based SSL method, the contrastive-based SSL method, and the integration of these two SSL methods. The experimental results of ablation study are shown in Table VIII.

Comparing Tables VII and VIII, we find that high classification accuracy can be achieved when there is only masking mechanism or contrastive-learning mechanism in the SSL method. For all datasets used in our experiments, the OAs of the contrastive-learning mechanism are higher than those of the masking mechanism, which is consistent with the phenomenon mentioned in the introduction of our paper. Furthermore, it is worth mentioning that the integration of the two mechanisms has obviously brought about a greater improvement. For example, the OA of the masking mechanism and the contrastive-learning mechanism on the Indian Pines dataset is 86.51% and 87.53%, respectively, while the OA

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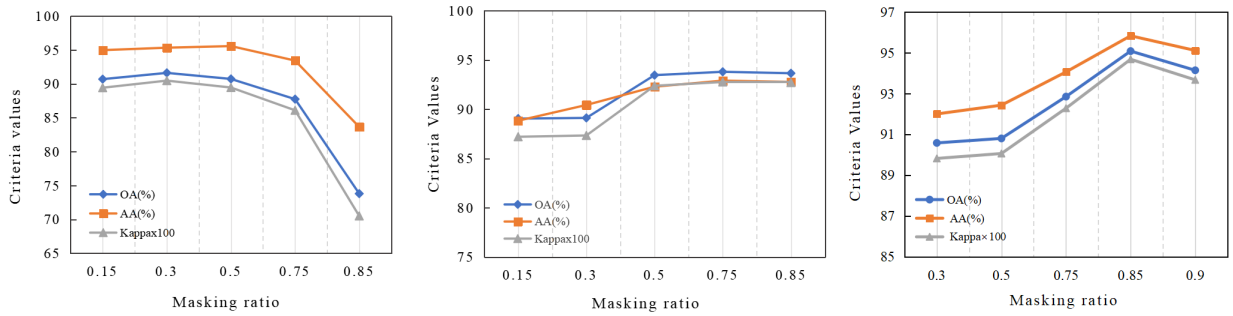
TABLE VIII
ABLATION STUDY RESULTS ON THREE DATASETS

Data Set	Criterion	2S3M+SFEM	PCLM+SFEM	MECL
Indian Pines	OA(%)	86.51	87.53	**91.62**
	AA(%)	93.71	93.64	**95.34**
	Kappa×100	84.76	85.86	**90.48**
WHU-Hi-HanChuan	OA(%)	91.26	92.81	**93.80**
	AA(%)	90.55	91.83	**92.90**
	Kappa×100	89.81	91.62	**92.77**
Houston University 2013	OA(%)	87.36	89.97	**95.08**
	AA(%)	89.15	91.54	**95.83**
	Kappa×100	86.34	89.16	**94.68**



(a) (b) (c)

Fig. 10. Sensitivity curves of masking ratios. (a) Indian Pines. (b) WHU-Hi-HanChuan. (c) Houston University 2013.

of the MECL on the Indian Pines dataset has even reached 91.62%. Specifically, the OA of MECL on the Houston University 2013 dataset is at least 5% higher than that of the other two mechanisms. This indicates that the contrastive-learning mechanism can effectively support the implicit reconstruction process of the masking mechanism. The similar results can be observed on the WHU-Hi-HanChuan dataset, demonstrating the significant advantages of the fusion mechanism. In summary, the ingenious fusion of the generative-based SSL method and the contrastive-based SSL method benefits HSIC more.

F. Sensitivity Analysis of Masking Ratios

In the previous exploration of classification experiments, the impact of masking ratio on the performance of MECL is easy to be observed. Therefore, we design specific experiments to analyze the sensitivity of masking ratios, the results of which are shown in Fig. 10.

Fig. 10(a) presents the sensitivity analysis results on the Indian Pines dataset, where MECL achieves the best classification accuracy at the masking ratio of 0.30. When the masking ratio is greater than 0.30, the performance of MECL shows a sharp downward trend. The OA has decreased by 13.94% when the masking ratio is adjusted from 0.75 to 0.85, indicating that MECL is sensitive to the masking ratio. The possible reason is that the total size of the Indian Pines dataset is relatively small for data-hungry deep networks. When the masking ratio is high, the remaining number of samples is insufficient to support the adequate learning of MECL.

Fig. 10(b) shows the sensitivity analysis results on the WHU-Hi-HanChuan dataset, the optimal masking ratio of which is 0.75. It is worth noting that when the masking ratio touches only 0.50, the three indicators have been already high. There is a certain delay in the optimal masking ratio compared with the Indian Pines dataset. The reason for the delay may be that the larger dataset requires more difficult reconstruction tasks. When the masking ratio is larger than 0.75, the performance of MECL begins to deteriorate, since the excessive masking ratio leads to severe information loss.

Fig. 10(c) shows the sensitivity analysis results on the Houston University 2013 dataset, the optimal masking ratio of which is 0.85. Interestingly, when the masking ratio is less than 0.85, the upward trend of OA on this dataset is very obvious with the increase of masking ratio. This phenomenon is opposite to that of the Indian Pines dataset. When the masking ratio is greater than 0.85, OA starts to decline. High masking ratio brings high OA and helps to reduce computational overhead, which is of great benefit for SSL that requires a large number of unsupervised samples.

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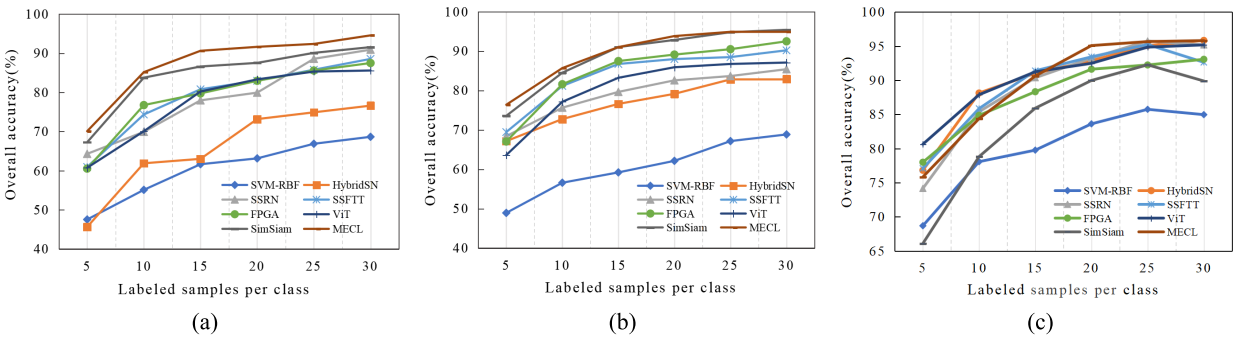

Fig. 11. OA curves of different numbers of labeled samples. (a) Indian Pines. (b) WHU-Hi-HanChuan. (c) Houston University 2013.

G. Impact of Different Numbers of Labeled Samples

Although the experimental condition of taking 20 labeled samples per class is already harsh, there is still room for exploration. Therefore, we randomly select 5, 10, 15, 20, 25, and 30 labeled samples from each category of three datasets for fine-tuning, so as to further prove the prominent performance of MECL with fewer labeled samples. Fig. 11 shows the OA curves of all mentioned methods under different numbers of labeled samples, where the performances of all methods are improved with the increase of labeled samples. As expected, the performance of MECL is more outstanding than that of other baselines on the first two datasets [Fig. 11(a) and (b)]. When the number of labeled samples is equal to or greater than 20, MECL outperforms all other methods on the Houston University 2013 dataset [Fig. 11(c)]. When the number of labeled samples is less than 20, the performance of MECL is very general. The possible reason is that all datasets are analyzed with the respective optimal masking ratio obtained from the sensitivity analysis of masking ratios experiments. The optimal masking ratio corresponding to the Houston University 2013 dataset is up to 0.85. When the number of labeled samples is relatively small, a high masking ratio can lead to performance degradation in MECL.

In short, all experimental results in Fig. 11 fully demonstrate the stability and reliability of our MECL.

V. Conclusion

In general, we propose a hybrid SSL method, that is, MECL. On the one hand, the proposed method is able to combine the idea of MAEs and the characteristics of hyperspectral data for effectively denoising, as well as fully extract local spatial information between adjacent pixels. On the other hand, our MECL utilizes the prototypical contrastive mechanism to make the abstract representations more consistent with the distribution of the original HSI data. The experimental results show that MECL with high stability, reliability, and generalizability is conducive to the HSIC.

Although the classification results are satisfactory, the pretext tasks are relatively complex and, therefore, challenging to implement. It is advisable to further explore simpler and effective pretext task. In addition, MECL requires a large amount of unlabeled data for training, which is a common issue with SSL algorithms, leading to additional computational overhead. This issue can be further improved through the application of efficient data preprocessing techniques and model compression techniques. Furthermore, how to better integrate generative-based and contrastive-based SLL methods remains a emerging topic worthy of in-depth discussion and study in HSIC.

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